Lab 1.3. Domain Name System (DNS)

Objectives

In this lab, we will use client DNS tools to explore the service structure on the Internet. Then, we will configure a name service based on BIND. The aim is to study both the basic configuration steps of the service and the protocol operation.

> On each exercise, provide the **commands used and their output**, the **screenshots of Wireshark captures performed**, and the **information specifically requested**.
>
> Enable **bi-directional shared clipboard** in the VMs (Devices menu).
>
> Use VirtualBox (View menu) to take **screenshots**.
>
> **Password** of user `cursoredes` is `cursoredes`.

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DNS Client

We will use client DNS tools, which are useful both to debug the deployment of a DNS service in our local network and to study the DNS structure in the Internet. The main tool to query DNS is `dig`. In this part, **we will use a physical machine**. If DNS queries to specific name servers are blocked, **use a web interface** like www.digwebinterface.com (enabling options "Stats" and "Show command").

Exercise 1. Check the contents of the DNS client configuration file, `/etc/resolv.conf`. Consult the `man` page of `resolv.conf` and study the meaning of options `nameserver` and `search`.

```
[cursoredes@localhost ~]$ cat /etc/resolv.conf
; generated by /usr/sbin/dhclient-script
search Home
nameserver 192.168.0.1
[cursoredes@localhost ~]$
```

If this file does not exist, only the name server on the local machine will be queried; the domain name is determined from the hostname and the domain search path is constructed from the domain name.

The different configuration options are:

nameserver Name server IP address
Internet address of a name server that the resolver should query, either an IPv4 address (in dot notation), or an IPv6 address in colon (and possibly dot) notation as per RFC 2373. Up to **MAXNS** (currently 3, see <*resolv.h*>) name servers may be listed, one per keyword. If there are multiple servers, the resolver library queries them in the order listed. If no **nameserver** entries are present, the default is to use the name server on the local machine. (The algorithm used is to try a name server, and if the query times out, try the next, until out of name servers, then repeat trying all the name servers until a maximum number of retries are made.)

domain Local domain name.
Most queries for names within this domain can use short names relative to the local domain. If no **domain** entry is present, the domain is determined from the local hostname returned by **gethostname**(2); the domain part is taken to be everything after the first '.'. Finally, if the hostname does not contain a domain part, the root domain is assumed.

search Search list for host-name lookup.
The search list is normally determined from the local domain name; by default, it contains only the local domain name. This may be changed by listing the desired domain search path following the *search* keyword with spaces or tabs separating the names. Resolver queries having fewer than *ndots* dots (default is 1) in them will be attempted using each component of the search path in turn until a match is found. For environments with multiple subdomains please read **options ndots:***n* below to avoid man-in-the-middle attacks and unnecessary traffic for the root-dns-servers. Note that this process may be slow and will generate a lot of network traffic if the servers for the listed domains are not local, and that queries will time out if no server is available for one of the domains.
The search list is currently limited to six domains with a total of 256 characters.

Exercise 2. Starting from root server `a.root-servers.net` and using the responses obtained, get the IP address of `informatica.ucm.es`. Complete the following table:

Name server	Name	TTL	Type	Data
a.root-servers.net	.es.	172800	NS	a.nic.es.
a.nic.es	.ucm.es.	86400	NS	chico.rediris.es.
chico.rediris.es.	informatica.ucm.es.	86400	CNAME	ucm.es.
	ucm.es.	86400	A	147.96.1.15

Note: Use command `dig @<server> <name> <type>`. More information in the man page of `dig`.

Exercise 3. Obtain the SOA record for `ucm.es.` using an authoritative server of the zone. Identify the relevant fields of the record.

dig SOA +additional +multiline ucm.es. @chico.rediris.es.

```
;; ANSWER SECTION:
ucm.es.                 86400 IN SOA ucdns.sis.ucm.es. hostmaster.ucm.es. (
                                2020102701 ; serial
                                28800      ; refresh (8 hours)
                                7200       ; retry (2 hours)
                                1209600    ; expire (2 weeks)
                                86400      ; minimum (1 day)
                                )
```

Exercise 4. Determine which mail server should be used to send an e-mail to `webmaster@fdi.ucm.es`, using an authoritative server of the zone.

dig MX +additional webmaster@fdi.ucm.es. @chico.rediris.es.

```
;; ANSWER SECTION:
webmaster\@fdi.ucm.es.  86400   IN      MX      1 aspmx.l.google.com.
webmaster\@fdi.ucm.es.  86400   IN      MX      5 alt1.aspmx.l.google.com.
webmaster\@fdi.ucm.es.  86400   IN      MX      5 alt2.aspmx.l.google.com.
webmaster\@fdi.ucm.es.  86400   IN      MX      10 aspmx2.googlemail.com.
webmaster\@fdi.ucm.es.  86400   IN      MX      10 aspmx3.googlemail.com.
webmaster\@fdi.ucm.es.  86400   IN      MX      10 ucsmtp.ucm.es.
```

Exercise 5. Determine the domain name for `147.96.85.71` starting from root server `a.root-servers.net` and usin the answers obtained. Complete the following table:

Name server	Name	TTL	Type	Data
a.root-servers.net	in-addr.arpa.	172800	NS	a.in-addr-servers.arpa.
a.in-addr-servers.arpa.	147.in-addr.arpa.	86400	NS	r.arin.net.
r.arin.net.	96.147.in-addr.arpa.	172800	NS	chico.rediris.es.
chico.rediris.es.	71.85.96.147.in-addr.arpa.	86400	PTR	www.fdi.ucm.es.

Note: Option `-x` of `dig` (in the web interface, it is enabled by selecting “Reverse” as record type) eases the reverse lookup when an IP address is detected as an argument, creating the search domain from the IP address (that is, reversing the byte order and adding `.in-addr.arpa.` domain) and establishing the default record type to PTR.

Exercise 6. Get the IP address of `www.google.com` using the default name server. Use option `+trace` of `dig` command (enabled with option “Trace” in the web interface) and observe the queries performed.

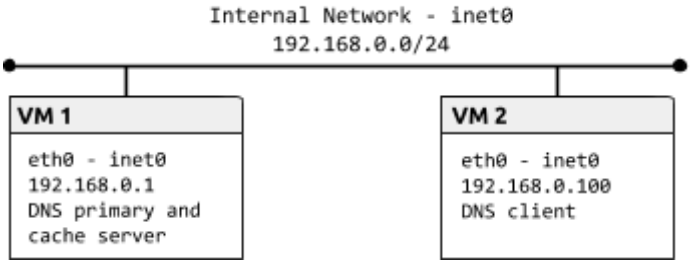
dig A +additional +trace www.google.com. @8.8.4.4

```
com.                    172800  IN      NS      m.gtld-servers.net.
com.                    172800  IN      NS      i.gtld-servers.net.
com.                    172800  IN      NS      h.gtld-servers.net.
com.                    172800  IN      NS      k.gtld-servers.net.
com.                    172800  IN      NS      l.gtld-servers.net.
com.                    172800  IN      NS      b.gtld-servers.net.
com.                    172800  IN      NS      e.gtld-servers.net.
com.                    172800  IN      NS      j.gtld-servers.net.
com.                    172800  IN      NS      f.gtld-servers.net.
com.                    172800  IN      NS      d.gtld-servers.net.
com.                    172800  IN      NS      c.gtld-servers.net.
com.                    172800  IN      NS      a.gtld-servers.net.
com.                    172800  IN      NS      g.gtld-servers.net.
;; Received 492 bytes from 192.36.148.17#53(192.36.148.17) in 56 ms

google.com.             172800  IN      NS      ns2.google.com.
google.com.             172800  IN      NS      ns1.google.com.
google.com.             172800  IN      NS      ns3.google.com.
google.com.             172800  IN      NS      ns4.google.com.
;; Received 280 bytes from 192.31.80.30#53(192.31.80.30) in 14 ms

www.google.com.         300     IN      A       216.58.192.196
;; Received 48 bytes from 216.239.38.10#53(216.239.38.10) in 11 ms
```

DNS Server

Environment Preparation

For this part, we will configure the network topology shown in the following figure:



As in previous labs, we will build the topology with the `vtopol` tool and an appropriate topology file. Configure the network interfaces as shown in the figure and check the connectivity between both VMs.

Forward Zone

VM1 will be the name server of the `labfdi.es` domain. Most resource records are defined in the forward zone.

Exercise 7. Configure the name server by adding a `zone` definition for the *forward* zone in file `/etc/named.conf`. The server `type` should be `master` and the zone `file` should be `db.labfdi.es`.

```
VM1
$man named.conf
$sudo nano /etc/named.conf

zone "labfdi.es." {
  type master;
  file "db.labfdi.es";
};

$sudo named-checkconf
```

Name

named.conf - configuration file for named

Synopsis

named.conf

Description

named.conf is the configuration file for **named**. Statements are enclosed in braces and terminated with a semi-colon. Clauses in the statements are also semi-colon terminated. The usual comment styles are supported:

C style: /* */

C++ style: // to end of line

Unix style: # to end of line

Acl

```
acl string { address_match_element; ... };
```

Key

```
key domain_name {
```

Check the default configuration and consult the `man` page of `named.conf` to learn the available options for the server and the zones. Recursion should be disabled on authoritative servers and queries should not be restricted (`allow-query` directive). Once the configuration file is created, execute the command `named-checkconf`, to verify that the syntax is correct.

Exercise 8. Create the file for the forward zone in `/var/named/db.labfdi.es` with the records specified in the following table. Specify also the `$TTL` command.

```
VM1
$sudo nano  /var/named/db.labfdi.es
$sudo named-checkzone labfdi.es /var/named/db.labfdi.es
$service named start
```

Record	Description
Start of Authority (SOA)	Freely choose the values for serial number, refresh, update, expiry and nx ttl. The primary name server is `ns.labfdi.es` and the contact e-mail is `contact@labfdi.es`.
Name Server (NS)	Name server is `ns.labfdi.es`, as specified in the SOA record
Mail server (MX)	Mail server is `mail.labfdi.es`
Canonical Name (CNAME) of server	`correo.labfdi.es` is an *alias* for `mail.labfdi.es`
Address (A) of name server	Address of `ns.labfdi.es` is `192.168.0.1` (VM1)
Addresses (A and AAAA) of web server	Addresses of `www.labfdi.es` are `192.168.0.200` and `fd00::1`
Address (A) of mail server	Address of `mail.labfdi.es` is `192.168.0.250`

Once the zone file is created, execute the command `named-checkzone <zone_name> <file>`, to verify that the syntax is correct. Finally, start the DNS service with command `service named start`.

Note: Don't forget that FQDN names must end with the root domain (`"."`). The zone name can be specified as `@` in the record name.

```
  GNU nano 2.3.1        File: /var/named/db.labfdi.es          Modified

$TTL 100000
labfdi.es.      IN      SOA     ns.labfdi.es. contact@labfdi.es. (
                        1234567         ; Serial
                             1h         ; Refresh
                            15M         ; Retry
                          1d10h         ; Expire
                          2h35M         ; Negative Cacha TTL
                        )
                IN      NS      ns.labfdi.es.
                IN      MX      mail.labfdi.es.
ser.labfdi.es.  IN      CNAME   mail.labfdi.es.
ns.labfdi.es.   IN      A       192.168.0.1
www.labfdi.es.  IN      A       192.168.0.200
www.labfdi.es.  IN      AAAA    fd00::1
mail.labfdi.es. IN      A       192.168.0.250


                        [ Read 17 lines ]
^G Get Help  ^O WriteOut  ^R Read File^Y Prev Page^K Cut Text  ^C Cur Pos
^X Exit      ^J Justify   ^W Where Is ^V Next Page^U UnCut Tex^T To Spell
```

```
[cursoredes@localhost ~]$ sudo named-checkzone labfdi.es /var/named/db.l
bfdi.es
zone labfdi.es/IN: loaded serial 1234567
OK
[cursoredes@localhost ~]$
```

Exercise 9. Configure the client VM to use the new name server. For this, create or modify `/etc/resolv.conf` with the new values for `nameserver` and `search`.

; generated by /usr/sbin/dhclient-script
domain ns.labfdi.es
search ns.labfdi.es
nameserver 192.168.0.1

Exercise 10. Use `dig` from the client to obtain information of the domain `labfdi.es`.

VM1
$dig labfdi.es

```
[cursoredes@localhost ~]$ dig labfdi.es

; <<>> DiG 9.9.4-RedHat-9.9.4-61.el7_5.1 <<>> labfdi.es
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 46034
;; flags: qr aa rd; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;labfdi.es.                     IN      A

;; AUTHORITY SECTION:
labfdi.es.              9300    IN      SOA     ns.labfdi.es. contact\@la
bfdi.es. 1234567 3600 900 122400 9300

;; Query time: 0 msec
;; SERVER: 192.168.0.1#53(192.168.0.1)
;; WHEN: Tue Nov 03 18:23:20 CET 2020
;; MSG SIZE  rcvd: 92

[cursoredes@localhost ~]$
```

Exercise 11. Perform more queries and, with the help of Wireshark:

- Check the protocol and port used by DNS client and server.
- Analyze the format (fields included and length) of the messages corresponding to DNS queries and responses.

VM2
$dig labfdi.es

No.	Time	Source	Destination	Protoc	Lengl	Info
1	0.00000000	192.168.0.100	192.168.0.1	DNS	80	Standard query 0xe1e9 A labfdi.es
2	0.00027628	192.168.0.1	192.168.0.100	DNS	134	Standard query response 0xe1e9
3	5.01435137	CadmusCo_4c:af:03	CadmusCo_ba:6b:75	ARP	42	Who has 192.168.0.100? Tell 192.168.0.1
4	5.01462384	CadmusCo_ba:6b:75	CadmusCo_4c:af:03	ARP	60	Who has 192.168.0.1? Tell 192.168.0.100
5	5.01466037	CadmusCo_4c:af:03	CadmusCo_ba:6b:75	ARP	42	192.168.0.1 is at 08:00:27:4c:af:03
6	5.01502068	CadmusCo_ba:6b:75	CadmusCo_4c:af:03	ARP	60	192.168.0.100 is at 08:00:27:ba:6b:75

```
[Response In: 2]
Transaction ID: 0xe1e9
▸ Flags: 0x0120 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 1
▾ Queries
  ▾ labfdi.es: type A, class IN
      Name: labfdi.es
      Type: A (Host address)
      Class: IN (0x0001)
▸ Additional records
```

Reverse Zone

Also, the server will include a database for reverse lookup. The reverse zone contains the PTR records corresponding to IP addresses.

Exercise 12. Add another zone definition for the reverse zone `0.168.192.in-addr.arpa.` in `/etc/named.conf`. The server type should be `master` and the zone `file` should be `db.0.168.192`.

```
VM1
$sudo nano /etc/named.conf.

zone "0.168.192.in-addr.arpa" {
  type master;
  file "db.0.168.192";
};
```

Exercise 13. Create the file for the reverse zone in `/var/named/db.0.168.192` with the SOA, NS and PTR records. This zone will use the same name server and configuration parameters in the SOA record. After that, restart the DNS service with command `service named restart` (or reload the configuration with command `service named reload`).

```
$TTL 100000
0.168.192.in-addr.arpa. IN SOA ns.labfdi.es. contact@labfdi.es. (
                              3         ; Serial
                         100000         ; Refresh
                          10000         ; Retry
                        1000000         ; Expire
                          10000 )       ; Negative Cache TTL
;
@                     IN      NS      ns.labfdi.es.
@                     IN      PTR     ns.labfdi.es.
1                     IN      PTR     ns.labfdi.es.
200                   IN      PTR     www.labfdi.es.
250                   IN      PTR     mail.labfdi.es.
```

Exercise 14. Check the reverse lookup by getting the name associated to address `192.168.0.250`.

VM1
$dig A -x `192.168.0.25`

```
[cursoredes@localhost ~]$ dig A -x  192.168.0.25

; <<>> DiG 9.9.4-RedHat-9.9.4-61.el7_5.1 <<>> A -x 192.168.0.25
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 23148
;; flags: qr aa rd; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;25.0.168.192.in-addr.arpa.     IN      A

;; AUTHORITY SECTION:
0.168.192.in-addr.arpa. 504870  IN      SOA     ns.labfdi.es. contact\@la
bfdi.es. 3 624890 75300 2412940 504870

;; Query time: 1 msec
;; SERVER: 192.168.0.1#53(192.168.0.1)
;; WHEN: Tue Nov 03 18:57:56 CET 2020
;; MSG SIZE  rcvd: 117
```